

PHYSICS 7B

UC BERKELEY

SPRING 2026

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## 7B Discussion Problems, Week 14

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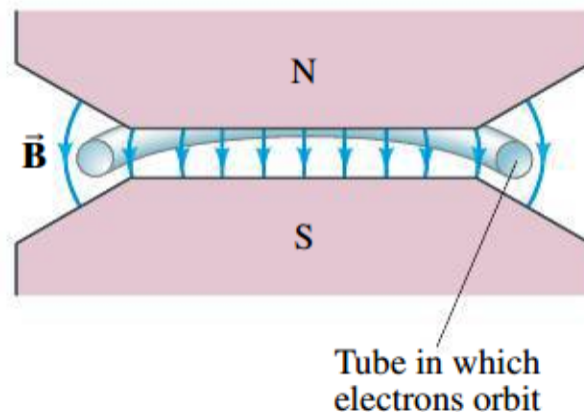
*Topics*

Electromagnetic induction and inductance

## Week 14 Discussion Problems

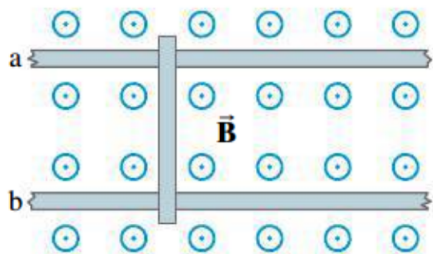
### Electromagnetic induction and inductance

- When a car drives through the Earth's magnetic field, an emf is induced in its vertical 75.0-cm-long radio antenna. If the Earth's field ( $5.0 \times 10^{-5} \text{ T}$ ) points north with a dip angle of  $45^\circ$ , what is the maximum emf induced in the antenna and in which direction(s) will the car be moving to produce this maximum value? The car's speed is 30.0 m/s on a horizontal road.
- The betatron, a device used to accelerate electrons to high energy, consists of a circular vacuum tube placed in a magnetic field (as shown in the figure below), into which electrons are injected. The electromagnet produces a field that (1) keeps the electrons in their circular orbit inside the tube, and (2) increases the speed of the electrons when  $B$  changes.
  - Explain how the electrons are accelerated.
  - In what direction are the electrons moving in the figure (give directions as if looking down from above)?
  - Should  $B$  increase or decrease to accelerate the electrons?
  - The magnetic field is actually 60 Hz ac; show that the electrons can be accelerated only during  $\frac{1}{4}$  of a cycle ( $\frac{1}{240} \text{ s}$ ). (During this time they make hundreds of thousands of revolutions and acquire very high energy.)
  - Show that the electrons in a betatron are accelerated at constant radius if the magnetic field  $B_0$  at the position of the electron orbiting the tube is equal to half the average value of the magnetic field ( $B_{\text{avg}}$ ) over the area of the circular orbit at each moment:  $B_0 = \frac{1}{2} B_{\text{avg}}$ . (This is the reason the pole faces have a rather odd shape in the figure.)

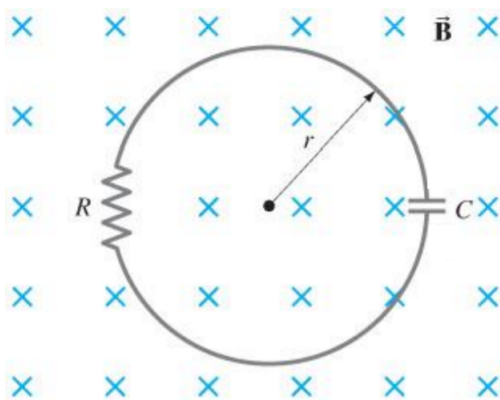


- A conducting rod (mass  $m$ , resistance  $R$ ) rests on two long frictionless and resistanceless parallel rails in a magnetic field  $\mathbf{B}$  (perpendicular to the rails and rod) as in the figure below.
  - If the rails are horizontal and the rod is given an initial push, will the rod travel at constant speed even though a magnetic field is present?
  - Suppose at  $t = 0$ , when the rod has speed  $v = v_0$ , the two rails are connected electrically by a wire from point a to point b across a resistor  $R$ . Determine the speed of the rod as a function of time. Discuss your answer. Assume the rails are a distance  $\ell$  apart.

Now suppose the two rails are a distance  $\ell$  apart, and that the magnetic field is uniform. At  $t = 0$ , the rod is at rest and a source of emf is connected to the points a and b. Determine the speed of the rod as a function of time if



- (c) the source puts out a constant current  $I$ ,
  - (d) the source puts out a constant emf  $\mathcal{E}_0$ .
  - (e) Does the rod reach a terminal speed in either case? If so, what is it?
4. (Bordel Final 2013) A circular-shaped circuit of radius  $r$ , containing a resistor of resistance  $R$  and a capacitor of capacitance  $C$ , is placed in a uniform magnetic field  $\vec{B}$ , pointing into the page. The capacitor is initially uncharged, and at time  $t = 0$ , the magnitude of the magnetic field starts changing at a rate  $dB/dt = a$ ,  $a > 0$ .
- (a) Using Lenz's law, determine the direction of the induced current in the closed loop. Justify your answer.
  - (b) Calculate the induced emf  $\epsilon$  and draw the emf symbol in the appropriate direction on the closed loop.
  - (c) What is the voltage  $V_c$  across the capacitor as a function of time?
  - (d) Draw the equivalent circuit right after  $t = 0$  and express the power dissipated by the resistor at  $t = 0$ .

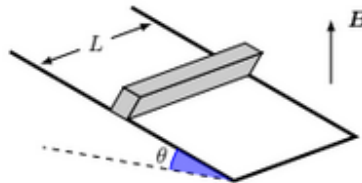
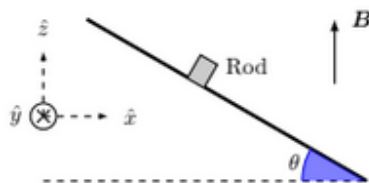


5. What is the energy dissipated as a function of time in a circular loop of  $N$  turns of wire having a radius of  $r$  and a resistance of  $R$  if the plane of the loop is perpendicular to a magnetic field given by

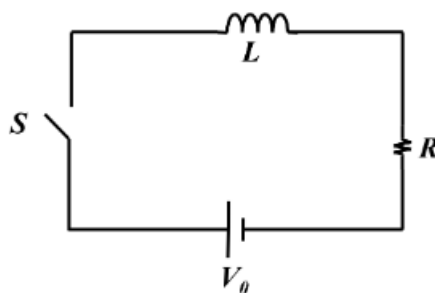
$$\mathbf{B}(t) = \mathbf{B}_0 e^{-t/\tau}?$$

6. Problem:

**Problem 21.2:** A rod with length  $\ell$ , mass  $m$ , and resistance  $R$  slides without friction down parallel conducting rails of negligible resistance, as shown below. The plane of the rails makes an angle  $\theta$  with the horizontal, and a uniform vertical magnetic field  $\mathbf{B}$  exists throughout the region.



1. Find the steady-state terminal velocity of the sliding rod.
  2. Show that the rate at which the internal energy of the rod is increasing is equal to the rate at which the rod is losing gravitational potential energy.
  3. Discuss the situation if  $\mathbf{B}$  were directed down instead of up.
7. LR circuit:  
Consider the LR circuit shown in the figure below.



- (a) (5 pts) Derive the expression for the current as function of the battery potential  $V_0$ , resistance  $R$ , time constant  $\tau$ , and time  $t$  after closing the switch at  $t = 0$ . Show and explain all your steps.

At  $t = 0$ , a  $10V$  battery is connected in series with a  $0.1H$  inductor and a total of  $10\Omega$  resistance.

- (b) (2 pts) What is the current at  $t = 0$ , i.e. when the switch is closed?
- (c) (2 pts) What is the time constant  $\tau$ ?
- (d) (2 pts) What is the maximum current?
- (e) (2 pts) At the maximum current, at what rate is energy being delivered by the battery?
- (f) (2 pts) At the maximum current, at what rate is energy being stored in the inductor's magnetic field?